

Time Series Data

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- Observations collected over a seq of time intervals
- daily, weekly, seconds, monthly,

* Characteristics

- Chronological order (regular interval or separated)
- Sequential orders
- temporal components (trends, seasonality, cycle, noise)
- constant frequency (continuous data without any missing value)
- Dynamic Nature (affected by external factors)

* Time Series Analysis

Statistical technique : meaningful insights about patterns & trend

- understanding past
- forecasting future

Most uses

line chart

For stock → candle stick

* Time Series Decomposition (breaking a time series into smaller understandable parts)

- Trend :
 - ↳ shows the overall long term direction of the data
- seasonality :
 - ↳ A pattern that repeats at a fixed interval (yearly, monthly, weekly)

Ex: Retail sales increases every Dec due to holidays

- Cyclic components:

↳ long term fluctuations that do NOT occur at fixed intervals.

② Usually caused by:

- economy, market conditions, business cycles

Ex: Bank profit rising during economic growth & fall during recession.
[unlike seasonality]

- Residual / Noise:

↳ random unexplained part left after removing trend & seasonality.

These are

- sudden spikes
- random events
- unpredictable behaviour

Types of Decompositⁿ

1. Additive:

- each component is added together
- seasonal variatⁿ remains roughly constant

$$Y = T + S + C + R$$

Original data Trend Seasonality Cyclic Residual/Noise

Intuition:

Suppose a store sells:

January	100
June	150
December	200

Every december, sales increase by +50
Even after years

Year	Normal Sales	Dec sales
2022	200	250
2023	300	350

The increase remains almost constant (+50)

{ Trend line + Constant seasonal waves + Random Noise

2. Multiplicative Decompositⁿ

- components multiply together
- seasonal fluctuations increase/decrease with trend

$$Y = T \times S \times C \times R$$

Imagine an e-commerce company growing rapidly.

Year	Avg Sales	Festival Sales
2022	1000	1500
2023	5000	8000
2024	20000	35000

Seasonal increase becomes larger every year
seasonal effect depends on the trend.

Stationarity

- statistical properties of data stay consistent over time.

- properties includes

 - o mean
 - o variance
 - o pattern

[a stationary time series behaves similarly throughout time]

- Constant Mean

avg remains similar over time

- Constant Variance

spread of data remains stable

- Constant Autocorrelatⁿ

relationship betⁿ current & past values remains stable.

1. Strict Stationarity

the entire probability distributⁿ remains unchanged over time

i.e. constant mean, variance & all higher moments

Ex: If you take

- 100 first observatⁿ

- 100 next observatⁿ

their distributⁿ should look identical

2.

Weak Stationarity

A series is weakly stationary if :

Mean is constant

$$E(X_t) = \mu$$

Variance is constant

$$\text{Var}(X_t) = \sigma^2$$

Covariance depends only on lag

- relationship depends on distance betⁿ points
not actual time

[Does not care about full distributⁿ]

1.

ADF Test (Augmented Dickey - Fuller Test)

used to check :

Is the time series stationary or non-stati...?

ADF checks whether the series has a :

- trend
- unit root

If unit root exists $\rightarrow$ data is non stationary

H_0 : data is non-stationary

H_1 : data is stationary

$p < 0.05 \rightarrow$ stationary

$p > 0.05 \rightarrow$ non-stationary

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2. KPSS Test (Kwiatkowski - Phillips - Schmidt - Shin Test)

also checks stationarity
BUT.....

Its hypothesis are opposite to ADF
checks, if trend is stationarity.

H_0 : Data is stationarity

H_1 : Data is non stationarity

$p < 0.5 \rightarrow$ Non stationary

$p > 0.5 \rightarrow$ stationary

Using both gives stronger confidence

* Interpretation :

ADF	KPSS	Meaning
Stationary	Stationary	Definitely stat.
Non Stationary	Non-stationary	Definitely Non stationarity

3. KS Test (Kolmogorov - Smirnov Test)

(Not for stationarity)

Do two datasets follow the same distributⁿ?

H_0 : both distributⁿ same

H_1 : both distributⁿ diff

$p < 0.05 \rightarrow$ Distributⁿ differ

$p > 0.05 \rightarrow$ Distributⁿ similar

White Noise & Random Walk

* White Noise

- completely random data with no pattern

constant mean & variance but no autocorrelation

Mean $E(X_t) = 0$

Constant Variance $\text{Var}(X_t) = \sigma^2$

No Correlation $\text{Cov}(X_t, X_{t-k}) = 0$, for all lags $k \neq 0$

* Residuals

Actual value - predicted values

A good forecasting model should produce residuals that behave like white noise because any remaining structure in residuals indicates that the model has failed to capture some underlying pattern in the data.

* Random Walk

- current value depends on previous value + random noise.

Ex: stock prices, cryptocurrency prices

* Ljung Box Test

Is there autocorrelation in the data ?

especially used on:

- residuals after forecasting

To verify whether residuals are :

- random
- independent